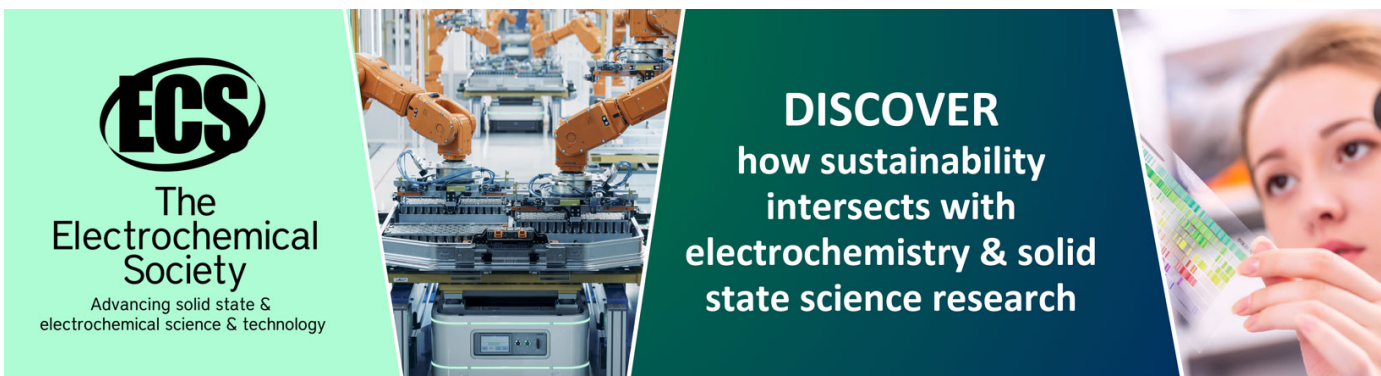


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Characterisation of Redlen HF-CdZnTe at $> 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ using HEXITEC_{MHz}

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ABSTRACT. In this paper, results are presented from the characterisation of Redlen Technologies high-flux-capable Cadmium Zinc Telluride (HF-CZT) hybridised to the HEXITEC_{MHz} ASIC, a novel 1 MHz continuous X-ray imaging system. A 2 mm thick HF-CZT HEXITEC_{MHz} detector was characterised on the B16 Test Beamline at the Diamond Light Source and displayed an average FWHM of 850 eV for monochromatic X-rays of energy 20 keV. Measurements revealed a shift in the baseline of irradiated pixels that results in a movement of the entire spectrum to higher ADU values. Datasets taken to analyse the effect's dynamics showed it to be highly localised and flux-dependent, with the excess leakage current generated equivalent to per-pixel shifts of $\sim 543 \text{ pA}$ (8.68 nA mm^{-2}) at a flux of $1.26 \times 10^7 \text{ ph s}^{-1} \text{ mm}^{-2}$. Comparison to results from a p-type Si HEXITEC_{MHz} device indicate this 'excess leakage-current' effect is unique to HF-CZT and it is hypothesised that it originates from trapping at the electrode-CZT interface and a temporary modification of the potential barrier between the CZT and metal electrode.

KEYWORDS: Hybrid detectors; Instrumentation for synchrotron radiation accelerators; Materials for solid-state detectors; X-ray detectors

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1 Introduction

Cadmium Zinc Telluride (CZT) is a compound semiconductor that has long drawn interest from the radiation-detection community. A high density ($\sim 5.8 \text{ g cm}^{-3}$) and resistivity ($\sim 10^{11} \Omega \text{ cm}$) make it highly suited for both the detection of high-energy X-rays and the fabrication of thicker detectors ($> 1 \text{ mm}$) unlike traditional detector materials (e.g. Si, CdTe) [1].

Despite these advantages, standard spectroscopic-grade CZT suffers poor performance at the high fluxes ($> 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$) demanded by medical applications such as Computed Tomography [2]. This is a result of poor hole-transport properties which lead to charge trapping and polarization under such conditions [3]. In response to this issue, Redlen Technologies has developed high-flux-capable CZT (HF-CZT) [4]. The growth method for this material optimises hole lifetimes, enabling its use in high-energy applications ($\leq 100 \text{ keV}$) at fluxes of $\leq 10^9 \text{ ph s}^{-1} \text{ mm}^{-2}$. Early lower-flux measurements were unable to identify any polarization in the material [5–7], but more recent synchrotron-based X-ray measurements indicate polarization-like phenomena at fluxes $> 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ [8].

The development of the HEXITEC_{MHz} spectroscopic X-ray imaging system allows the dynamics of high-flux phenomena to be investigate on a $< 1 \text{ ms}$ timescale. HEXITEC_{MHz} is the successor to the 2009 HEXITEC (High Energy X-ray Imaging Technology) ASIC, designed at the UKRI STFC Rutherford Appleton Laboratory (RAL) for the readout of high-Z sensors (CdZnTe, CdTe, GaAs:Cr) in hard-X-ray ($> 20 \text{ keV}$) spectroscopic-imaging applications [9, 10]. To date, HEXITEC has been deployed across a range of sectors that utilise imaging techniques such as Compton X-ray Imaging and Hyperspectral X-ray Tomography [11, 12]. Although HEXITEC continues to demonstrate the benefits of hyperspectral information, the 9.1 kHz maximum frame rate limits the spectroscopic X-ray flux. Frame occupancies $< 10 \%$ are required for energy-resolution-improving charge-sharing corrections,

restricting HEXITEC to spectroscopic fluxes $< 10^4 \text{ ph s}^{-1} \text{ mm}^{-2}$. The limitations of the original system mean there is a demand for a continuous X-ray detector with a faster readout.

HEXITEC_{MHz}, like its predecessor, provides fully-spectroscopic X-ray imaging on an 80×80 pixellated array. However, it bears several key differences to the original's design. Most notably, HEXITEC_{MHz} utilises an integrating Front-End, 12-bit in-pixel digitisation, and twenty 4.1 Gbps data-readout serialisers [13]. These enable a continuous 1 MHz frame rate, > 100 times faster than HEXITEC, and thereby allows for fluxes $> 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ in spectroscopic-imaging applications. Despite the technical challenges of delivering MHz-frame-rate imaging, early measurements indicate a similar spectroscopic performance to HEXITEC, with targeted full width at half maximum (FWHM) of $< 1 \text{ keV}$ at 100 keV in CZT [14–16]. Measurements using the HEXITEC_{MHz} ASIC currently utilise prototype detector hardware, with a productised camera system to be delivered in 2024.

To analyse the performance of 2 mm thick HF-CZT hybridised to HEXITEC_{MHz}, a prototype system was characterised on the Diamond Light Source (DLS) B16 Test Beamline. Results obtained using the uniform, monochromatic, and high-flux X-ray source are presented here.

2 HEXITEC_{MHz} detector system

2.1 HEXITEC_{MHz} ASIC

The HEXITEC_{MHz} ASIC is designed for fully-spectroscopic X-ray imaging at a continuous 1 MHz frame rate with electron-readout materials such as CZT, CdTe and p-type Si [13]. The ASIC is fabricated on a 180 nm TSMC complementary metal-oxide semiconductor (CMOS) process and comprises an 80×80 array of 6400 channels on a $250 \mu\text{m}$ pixel pitch. Two edges of the array are occupied by power and serialiser connections, resulting in a two-side buttable system with a maximum power consumption of $\sim 15 \text{ W}$.

Charge induced on a pixel pad is integrated within the pixel Front End by a charge-sensitive amplifier (CSA) that is connected in parallel to $\sim 7 \text{ fF}$ and $\sim 14 \text{ fF}$ feedback capacitors (C_{CSA1} , C_{CSA2}). These capacitors can be toggled by separate switches, enabling selection of one of three static dynamic ranges. These correspond to ranges of $\sim 100 \text{ keV}$, $\sim 200 \text{ keV}$ and $\sim 300 \text{ keV}$ in CZT. Stored charge is cleared using a reset switch and although the ASIC operates at a MHz rate, this reset can be impeded for multiple frames. This allows integration times of $\leq 255 \mu\text{s}$. Small variations in the CSA baseline occur following reset and a correlated double-sampling (CDS) amplifier removes these by sampling the CSA output twice, initially shortly after reset. To enable this, reset of the CDS amplifier is delayed relative to that of the CSA, resulting in a total reset time of $\sim 100 \text{ ns}$. The CDS output is stored on the sample-and-hold capacitor once per frame, ready for digitisation.

To enable a continuous MHz frame rate, digitisation occurs on chip in time-to-digital convertors (TDCs). Here, the sampled voltage acts as one input to a comparator, with the other provided by a dynamic signal from a ramp generator. This signal slews at a defined rate and the comparator fires when the ramp matches the sampled voltage. This stops an incrementing ripple counter (1.64 GHz clock) with the final 12-bit value proportional to the sampled signal. One ramp generator is shared between two pixel columns whilst the remaining TDC digital circuitry is shared within *super pixels*, groups of 2×4 pixels.

Digitised data is shifted down the pixel array to be output over the ‘fast-data stream’. The 80×80 array is subdivided into four-column divisions, each containing a packet assembler and dedicated

serialiser. The packet assembler converts the data from all four columns, four rows at a time, into three 64-bit data packets. These are encoded using the proprietary Xilinx[®] Aurora 64b/66b protocol, written into FIFOs (First In, First Out) and read out of the 4.1 Gb s⁻¹ serialisers. This results in a total raw-data rate of 10 GB s⁻¹.

The described process is pipelined with the integration of frame $N + 2$, digitisation of frame $N + 1$ and readout of frame N occurring simultaneously. Further details on the ASIC design are given in L. Jones et al., 2022 [13].

2.2 Data acquisition and detector control

On-board Samtec FireflyTM modules transfer digitised data over optical fibre to an Alpha-Data ADM-PCIE-9V5 PCI Express card containing the Xilinx[®] Virtex UltraScale Plus VU9P-3 FPGA. Here, Aurora-encoded data is decoded and sent over 100 Gbit Ethernet to a computer running ODIN data, a UKRI-STFC-developed scalable data-acquisition-and-processing network [17]. Data frames and the corresponding frame numbers are saved into a HDF5 file [18].

These measurements utilised a beta version of the FPGA firmware, allowing readout of a single serialiser channel. This corresponded to a 4×80 region of the pixel array. Any of the 20 channels could be selected however, and tests confirmed that all 20 Firefly-Alpha-Data links were active [13].

The productised system will also implement a Xilinx[®] Alveo U50 Data Center Accelerator card as a second-stage FPGA. This will provide real-time per-pixel corrections (dark subtraction, linearity correction etc.) and per-event processing (charge-sharing corrections, histogramming etc.), reducing the data rate by orders of magnitude.

Data acquisition and ASIC settings are controlled using the LOKI system, a Xilinx[®] Zynq system-on-chip (SOC) operating ODIN Control. LOKI communicates with HEXITEC_{MHz} over the Serial Peripheral Interface (SPI), writing to ASIC registers in 8-bit transactions (7-bit address, 1-bit value). These registers control the ASIC configuration and include user-defined settings such as the gain stage and integration time. They also configure the timing block, serialisers and TDCs. Optimising HEXITEC_{MHz} has involved establishing which register combinations provide the best spectroscopic performance.

PCB sensors also enable LOKI to monitor environmental conditions including temperature and humidity. The detector is mounted on a Peltier within a humidity-controlled enclosure and temperature information from LOKI is fed back into its power supply for maintenance of a stable operating temperature.

2.3 HF-CZT HEXITEC_{MHz} detector

Standard spectroscopic-grade CZT has been produced by Redlen Technologies since the early 2000s. Designed for lower-flux gamma-ray spectroscopy applications requiring thick detectors (> 2 mm), this material utilise a growth process optimised to deliver excellent electron charge-transport properties and spectral resolution [19]. However, this results in short hole carrier lifetimes ($\tau_h \approx 0.2$ μ s) and due to the poor hole mobilities inherent to CZT material ($\mu_h \approx 0.1 \mu_e$), this leads to significant hole trapping and polarization at high fluxes [3].

Since 2017, Redlen have also started to provided high-flux capable CZT (HF-CZT), which was developed for operation at the high X-ray fluxes ($\leq 10^9$ ph s⁻¹ mm⁻²) required for medical computed tomography [2]. For this material grade, the growth method is modified to prioritise the

hole-transport properties, resulting in an order-of-magnitude increase in hole lifetimes ($\tau_h \approx 2 \mu\text{s}$) and a high-flux-capable material [5].

The HF-CdZnTe material tested for this report was grown by Redlen Technologies using a proprietary travelling-heater growth method [4]. The completed sensors were also fabricated and processed by Redlen and comprise two Pt electrodes — a planar cathode and a pixelated anode of 80×80 pixels on a $250 \mu\text{m}$ pitch ($25 \mu\text{m}$ inter-pixel gap). A guard ring surrounds the pixelated array — $100 \mu\text{m}$ wide along three edges and $200 \mu\text{m}$ along the final edge. This results in total sensor dimensions of $20.35 \text{ mm} \times 20.35 \text{ mm} \times 2.00 \text{ mm}$. Equivalent sensors have been previously tested using the original HEXITEC ASIC, with a reported FWHM of $\sim 830 \text{ eV}$ at 59.54 keV [20].

A HF-CZT sensor was hybridised to HEXITEC_{MHz} by the UKRI STFC Interconnect team. Gold-studded ASIC pixels and silver-loaded-epoxy bump-bonded sensor pixels were flip-chip bonded (using a SET FC150) and cured in-situ (150°C), before ASIC I/O pads were Al-wedge-bonded to the PCB [21]. An image of the completed sensor, operated at a bias voltage of -1000 V (500 V mm^{-1}), is shown in figure 1.

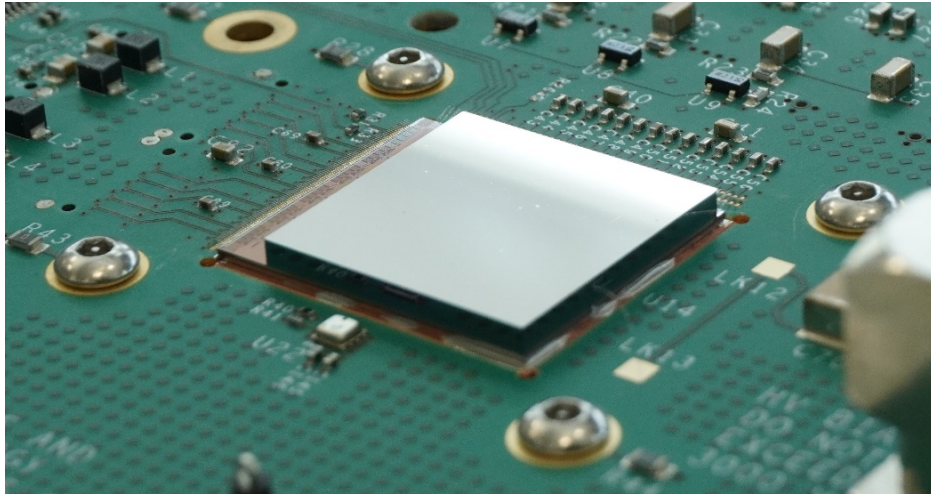


Figure 1. A 2 mm thick HF-CZT sensor hybridised to the HEXITEC_{MHz} ASIC.

3 Method

3.1 B16 test beamline experimental setup

Data was collected on the B16 Test Beamline at the DLS synchrotron using a water-cooled Si (111) double crystal monochromator with an energy resolution of $< 100 \text{ eV}$ at 10 keV ($\Delta E/E < 10^{-2}$) [22]. This provided monochromatic X-rays in the range of $12 \text{ keV} - 35 \text{ keV}$, with the flux at a particular energy controlled using a selection of aluminium and carbon attenuators.

The tested devices were mounted on an optics table with two sets of JJ-slits used to remove scatter and define the X-ray beam size between limits of $20 \mu\text{m} \times 20 \mu\text{m}$ and $3000 \mu\text{m} \times 3000 \mu\text{m}$. Devices were housed in a temperature-controlled die-cast enclosure and, unless stated, results were acquired at an ASIC temperature of 20°C . For operation below ambient conditions, two solid-state de-humidifiers on the enclosure's exterior prevented condensation forming on the ASIC.

Datasets were collected in all three gain stages and as discussed in section 2.2, comprise the output of a single serialiser channel (4×80 pixels). The incident flux was a function of both the beam energy and the attenuation used ($C \leq 1.5$ mm and $Al \leq 8$ mm). For spectroscopic characterisation, data was collected at frame occupancies of $< 10\%$ to ensure that, stochastically, events in adjacent pixels were a consequence of charge sharing. This allowed the application of charge-sharing corrections (section 3.2).

Unless stated otherwise, X-ray fluxes were estimated by integrating the total energy above the per-pixel threshold and dividing by the product of the beam energy (20 keV) and the total sample period (1 s). This value was subsequently multiplied by a factor of 10/9 to account for the 10 μ s reset period (10 % of total frame time). This correction is appropriate at lower occupancies as HEXITEC_{MHz} operates approximately as a nonparalyzable counting system with a dead time unaffected by events occurring within the ASIC's reset window. Note that this method ignores the presence of higher-order photons within the beam.

3.2 Data analysis

Bespoke Python scripts were used for the analysis of all collected data [23–26].

Prior to analysis, device calibrations were generated for each dataset to enable conversion between raw- (ADU) and energy- (keV) per-pixel spectra. A peak-fitting algorithm was used to identify both the noise (0γ) and primary harmonic (1γ) peaks, and a linear fit generated based on their ADU centroids and known energies. This resulted in per-pixel gradient and intercept coefficients and is shown for a single pixel within a 20 keV high-gain dataset in figure 2.

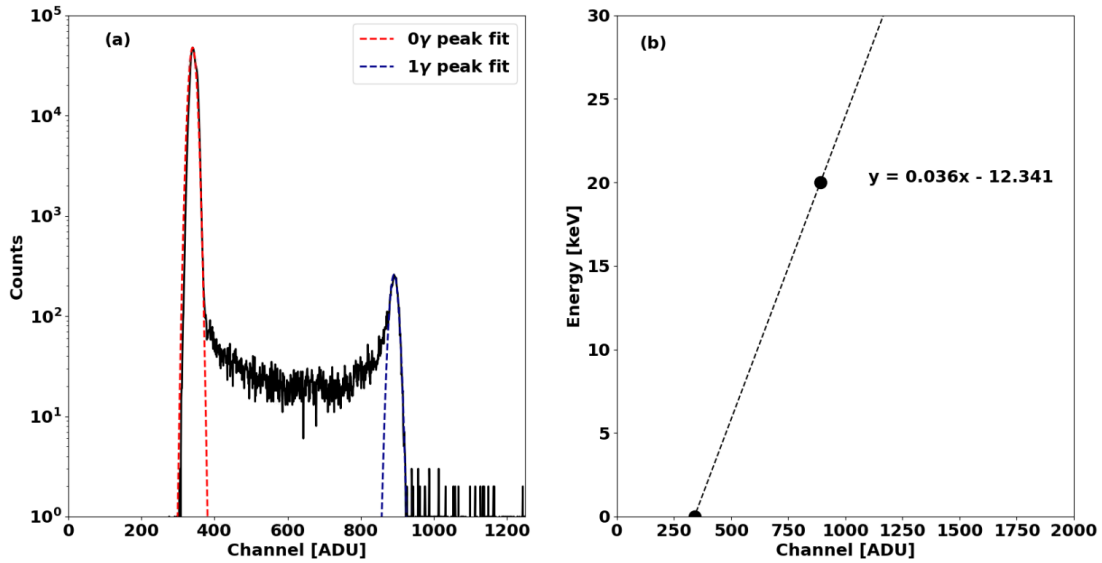


Figure 2. The calibration of pixel (13,37) within a high-gain 20 keV dataset. (a) The raw spectrum of pixel (13,37) showing Gaussian fits to the 0γ and 1γ (20 keV) peaks — 1 ADU bin width. (b) The linear calibration resulting from these fits.

The spectroscopic resolution of a detector can be improved by implementing charge-sharing corrections to events shared across neighbouring pixels (signals above a threshold in adjacent pixels within a 3×3 matrix). These corrections require frame occupancies $< 10\%$ (section 3.1) and one such technique is charge-sharing discrimination (CSD) [27]. CSD discards shared events, improving the resolution of

individual peaks through removal of a low-energy background continuum. The effects of CSD are shown in figure 3, where summed raw and CSD-corrected spectra are shown for a high-gain 20 keV dataset.

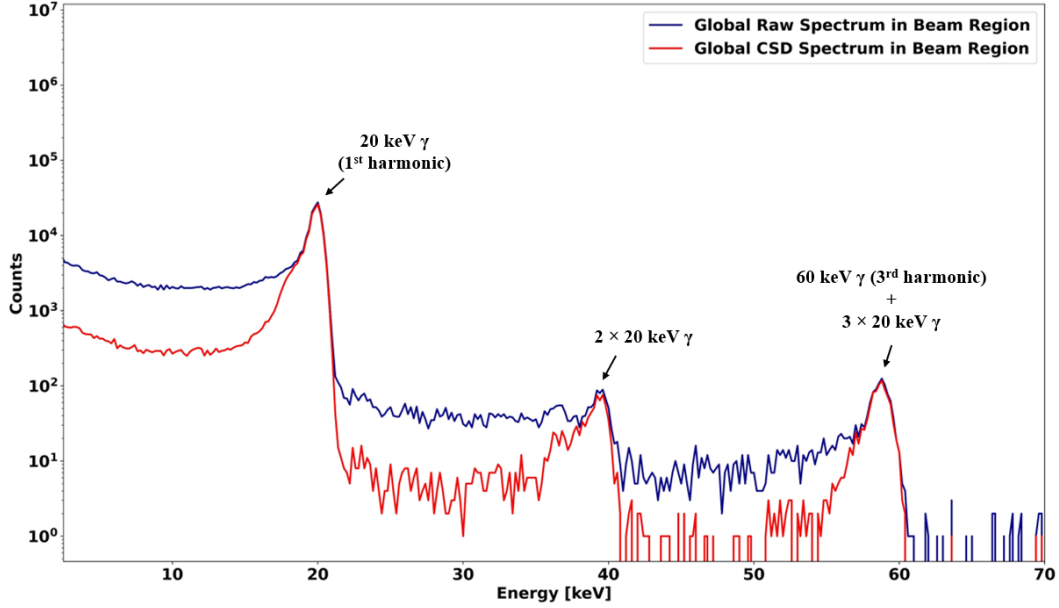


Figure 3. A comparison between the calibrated summed raw (blue) and CSD (red) spectra of illuminated pixels within a high-gain 20 keV dataset using 80 mm Al attenuation ($2.93 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$) - 0.1 keV bin width. Key spectroscopic features are labelled.

Note that low-energy thresholds must be calculated for each exposed pixel to differentiate event signals from electronic noise. For this report, thresholds 4σ above the 0γ peak’s centroid ($\mu + 4\sigma$) were chosen. This is equivalent to ~ 1 keV. Additionally, corrections cannot be applied to edge pixels as data from all 8 nearest neighbours is required. CSD analysis therefore only utilised the central two columns of the four-column readout from the single serialiser channel.

Key features of the X-ray beam can also be identified within this figure. Alongside the 20 keV photo peak, lower-intensity peaks are also identifiable at 60 keV and 40 keV. The former results from the generation of higher-order photons within the monochromator alongside the primary harmonic — 60 keV represents the 3rd harmonic. The intensity of this peak is > 100 times lower than the 20 keV peak and includes counts from the integration of three 20 keV photons interacting in both the same pixel and integration window. The integration of two 20 keV photons has resulted in the 40 keV peak as the 2nd harmonic is not transmitted by the monochromator. The deviation of the 3rd harmonic from 60 keV is a result of limitations in the calibration method and the poorer statistics available at higher energies.

4 Results

4.1 Spectroscopic imaging characterisation

The spectroscopic performance of the HF-CZT HEXITEC_{MHz} device was quantified through calculation of the primary-photo-peak FWHM at a beam energy of 20 keV and 1.83 % occupancy ($2.93 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$). Above this energy, X-ray fluxes from the double crystal monochromator rapidly decreased.

Per-pixel FWHM in the analysed 2×10 pixel region were calculated using a Gaussian fit to the 1γ peak in the CSD spectra. This employed the same peak-fitting algorithm used in calibration (section 3.2) and is shown for pixel (13,37) within the high-gain dataset in figure 4.

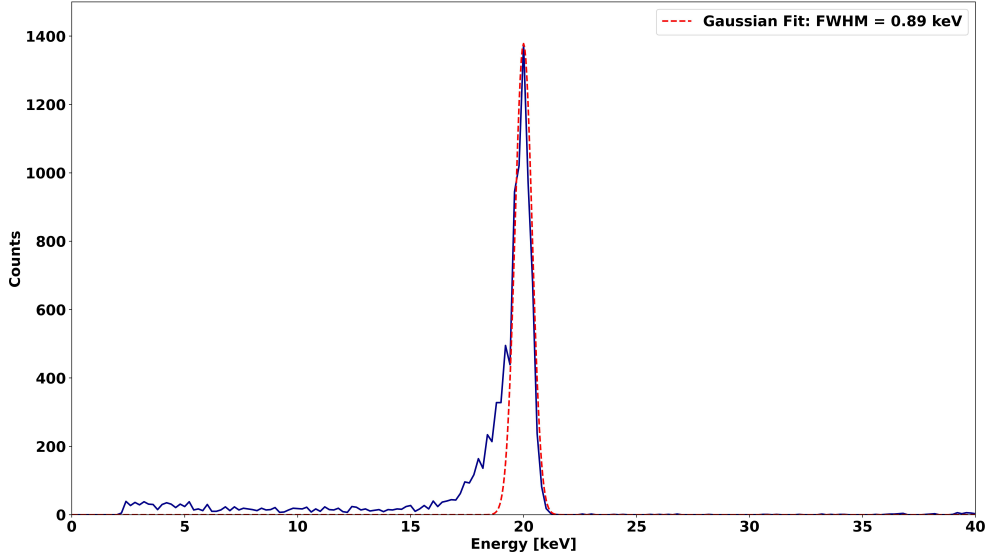


Figure 4. The calibrated high-gain CSD spectrum of pixel (13,37) showing a Gaussian fit to the 20 keV photopeak - 0.2 keV bin width.

The FWHM distributions obtained in each operating mode at 20 keV are shown in figure 5 alongside a summary of these results in table 1. The average FWHM of 850 eV and 920 eV for the high- and medium-gain stages exceed the targeted 1 keV spectroscopic resolution for continuous MHz imaging. For comparison, an average FWHM of 790 eV at 59.54 keV was measured using a HEXITEC ASIC hybridised to 2 mm thick Redlen HF-CZT material from a different wafer [20]. The HEXITEC Front End utilises a $2 \mu\text{s}$ shaper for improved noise performance and consequently operates at a maximum frame rate of 9.1 kHz. Additionally, datasets taken during the experiment at 15 keV using a 300 μm thick p-type Si HEXITEC_{MHz} device gave high- and medium-gain resolutions of 682 ± 57 eV and 814 ± 59 eV respectively [16]. Here, the superior p-type Si resolutions are expected on account of the relative electron-hole-pair generation energies of these two materials (3.62 eV for Si cf. 4.67 eV for CZT) [28].

Although the high- and medium-gain results lie within the targeted resolution, these values greatly exceed the 213 eV fundamental Fano-limited resolution for CZT at 20 keV (using a CZT

Table 1. A summary of the spectroscopic results at 20 keV. Errors quoted are the standard deviation in values obtained across the analysed 2×10 pixel regions.

Peak Energy [keV]	High-Gain FWHM [eV]	Medium-Gain FWHM [eV]	Low-Gain FWHM [eV]
20	850 ± 100	920 ± 110	1130 ± 120

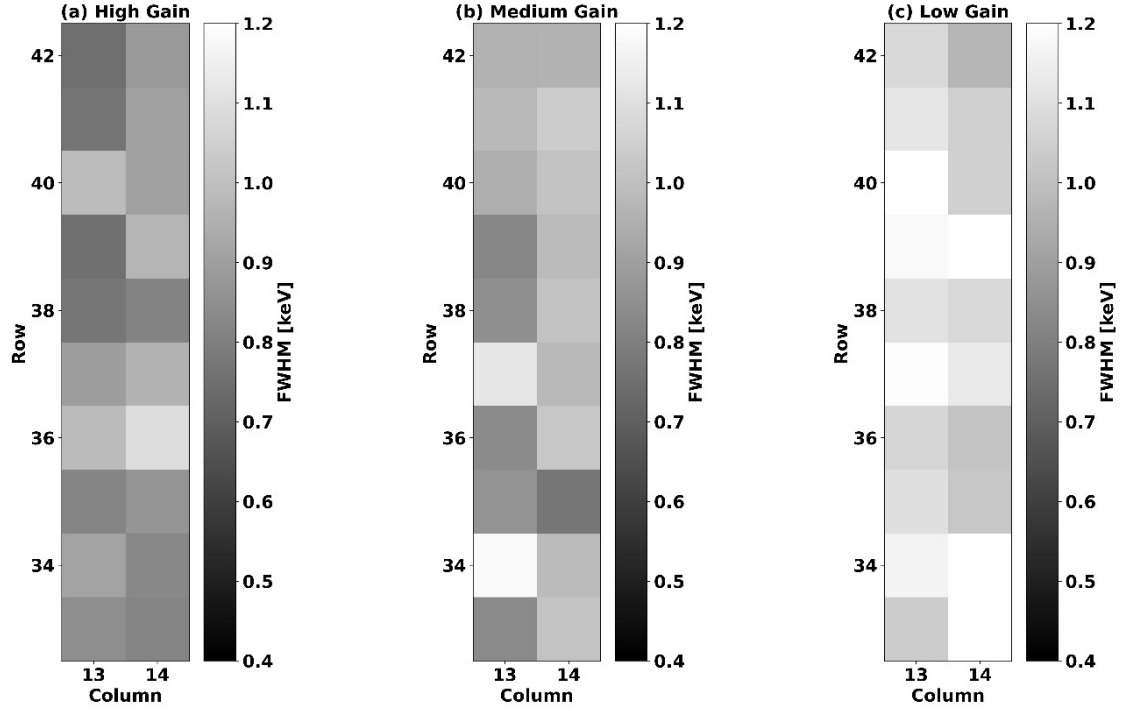


Figure 5. The FWHM distributions obtained at 20 keV across the analysed 2×10 pixel region. (a) High-gain results. (b) Medium-gain results. (c) Low-gain results.

Fano factor of 0.089) [29]. This limit results from stochastic variations in the number of charge carriers generated at a specified energy and can be used to approximate the ASIC’s electronic noise if material and electronic contributions to the overall device noise are assumed uncorrelated. This allows the summation of the individual components in quadrature, yielding an estimated electronic noise of ~ 820 eV in the high-gain stage.

Further optimisation of ASIC register settings may improve the resolutions obtained. Register values were chosen using results from the electrical characterisation of a bare ASIC and this process should be repeated using a hybridised device to account for additional noise contributions (e.g. sensor leakage current).

For these measurements only a single fast-data channel was readout, and it is therefore currently difficult to comment upon the variations in FWHM values measured across the analysed region. The delivery of the FPGA firmware required for a full 80×80 pixel readout will allow for a better assessment of performance variations and device uniformity across the active area.

4.2 Offset effect

4.2.1 Identification

As mentioned in section 3.2, a unique device calibration was generated for each dataset used in the above spectroscopic characterisation. This used a linear fit to the 0γ and 1γ peaks within the per-pixel CSD spectra and inherently corrected for each pixel’s leakage current. Here, the 0γ peak is representative of the thermally-induced charge on a pixel within a frame in the absence of an X-ray (often referred to as the leakage or dark current). In an ideal detector, the location of the 0γ peak’s

centroid should be identical in datasets taken in the absence of X-rays (dark datasets) and datasets in which the pixel is irradiated. This is indicative of a detector that fully recovers between frames, resulting in a response independent of the pixel's radiation history.

This ideal behaviour was not seen in the tested HF-CZT $\text{HEXITEC}_{\text{MHz}}$ device, and figure 6 compares a high-gain dark dataset to those collected at 20 keV using 80 mm and 55 mm of Al attenuation. This attenuation resulted in respective X-ray fluxes of $\sim 0.30 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (1.85 % occupancy) and $\sim 3.15 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (19.69 % occupancy). As shown in the uncorrected pixel (13,37) spectra of Fig 6a, the 0γ peak was shifted to higher ADU values upon X-ray illumination with a centroid shift of 35 ADU between the dark and $\sim 0.30 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ datasets. Comparison between irradiated datasets demonstrates that this offset was also dependent upon the X-ray flux. Relative to the $\sim 0.30 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ dataset, the centroid was shifted by a further 136 ADU upon an increase in X-ray flux of $\sim 2.85 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$.

This phenomenon was not unique to pixel (13,37) and shifts towards higher ADU values under X-ray illumination persist across the entirety of the illuminated 4×10 pixel region. This is shown in figure 6(b), which maps the difference in 0γ peak position between the dark and $\sim 3.15 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ datasets across the entire serialiser channel. This map and the associated plot of the average ADU shift within each readout row (Fig 6c) reveals negligible shifts (< 2 ADU) outside of the exposed detector area, providing an early indication that the effect was constrained to illuminated pixels.

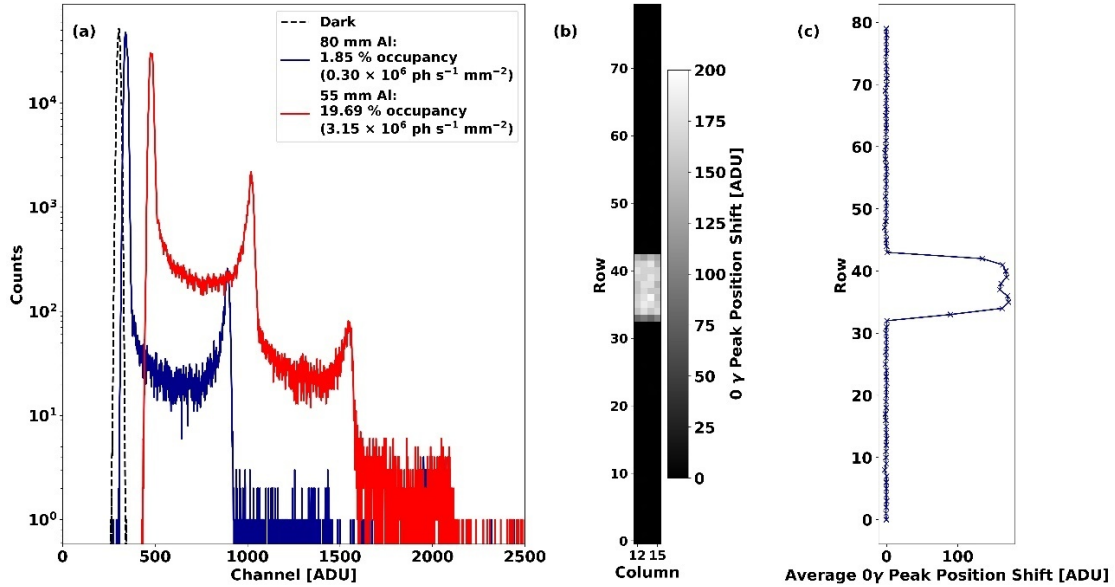


Figure 6. Results comparing 20 keV high-gain datasets collected at fluxes of $\sim 0.30 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (80 mm Al) and $\sim 3.15 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (55 mm Al). (a) The raw spectra obtained for pixel (13,37) — 1 ADU bin width. (b) Map of the shifts in the position of the 0γ peak of the 55 mm Al dataset relative to a dark dataset. (c) The average shift in the 0γ peak position measured in each readout row of the 55 mm Al dataset.

Importantly however, this behaviour was not observed in datasets taken at the B16 Test Beamline using a p-type Si $\text{HEXITEC}_{\text{MHz}}$ device. This is shown in figure 7, a comparison of a high-gain dark dataset to those collected at 10 keV under respective X-ray fluxes of $\sim 0.63 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (3.94 % occupancy) and $\sim 1.13 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (7.05 % occupancy). Here, the choice of 10 keV, compared to the higher-energy 20 keV HF-CZT datasets, was driven by the lower density and X-ray attenuation

of the p-type Si material. Across these datasets, the location of the 0γ peak in pixel (14,45) remained fixed and similar results are seen across the 4×80 readout in figure 7(b). Here, any small shifts of magnitude < 3 ADU were likely the result of minor ASIC power-supply fluctuations, causing variations in the slew rate of the ramp generator used during digitisation (section 2.1). The larger shifts seen for pixels (15,44) and (15,23) are the consequence of imperfections in the hybridisation process, resulting in their anomalous behaviour, and can be discounted.

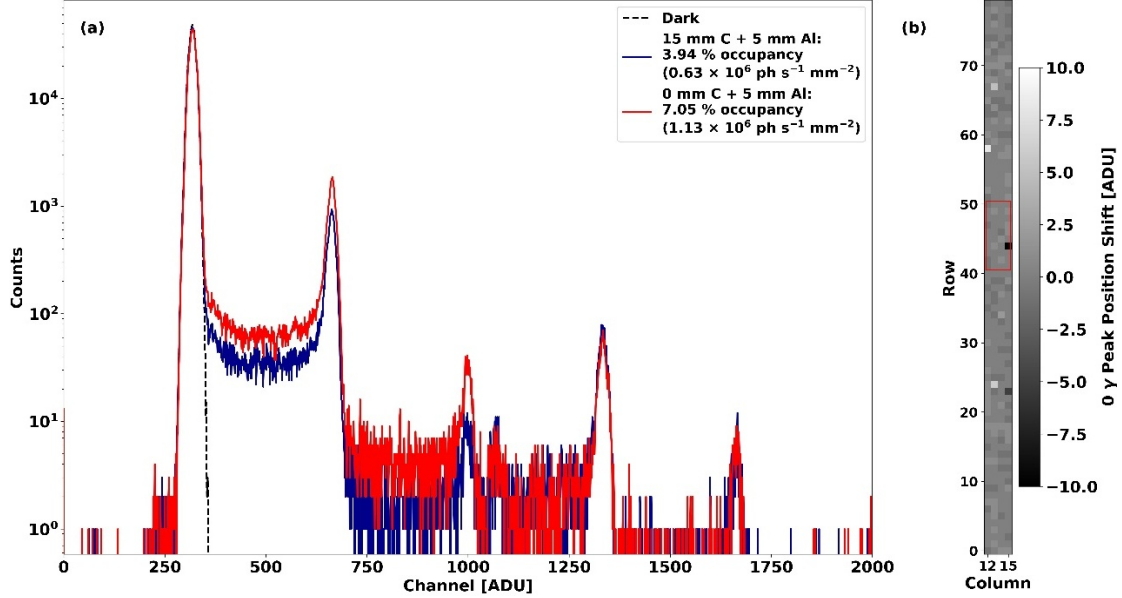


Figure 7. Results comparing 10 keV p-type Si HEXITEC_{MHz} high-gain datasets collected at fluxes of $\sim 0.63 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (15 mm C + 5 mm Al) and $\sim 1.13 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (0 mm C + 5 mm Al). (a) The raw spectra obtained for pixel (15,44) — 1 ADU bin width. (b) Map of the shifts in the position of the 0γ peak of the 5 mm Al dataset relative to a dark dataset. The position of the X-ray beam is highlighted in red.

The comparison of the HF-CZT and p-type Si results suggests that this phenomenon is not a HEXITEC_{MHz} ASIC effect, but rather a sensor effect. More precisely, the shift in the position of the 0γ peak is the result of excess leakage current generated within the detector following the initial application of a specific flux and is the result of specific material charge-trapping mechanisms.

Assuming a CZT atomic Zn content of 10 %, the penetration depth of 20 keV X-rays within the material is $\sim 80 \mu\text{m}$ [30]. Consequently, most of the incident flux at this energy interacts within a volume $\sim 100 \mu\text{m}$ below the pixelated cathode. Assuming a μ_h of $100 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, an applied bias voltage of -1000 V will result in the extraction of the generated holes within $\sim 1 \text{ ns}$ [5]. This is orders of magnitude faster than the measured hole lifetime ($\tau_h \approx 2.5 \mu\text{s}$) for the HF-CZT material, suggesting the phenomenon is not the result of hole trapping within the bulk CZT [5].

Instead, it is hypothesised that the observed phenomenon is the result of trapping at the electrode-CZT interface, with such effects previously studied in relation to inter-pixel conductivity [31]. The surface of the CZT and the interface to the metal electrode are known to be regions containing higher concentrations of defects and trapping centres [32], resulting in an accumulation of space charge that modifies the electrode-CZT potential barrier. This would manifest in a change in the observed pixel leakage current. Additional datasets, taken at different energies and temperatures, but identical flux, are required to test this hypothesis further.

For ease of reference, this effect will be referred to as the ‘excess leakage-current effect’ for the remainder of the paper. It is worth noting that a recent characterisation of the performance of HF-CZT using electroless gold and sputtered platinum electrodes at fluxes up to $10^{12} \text{ ph s}^{-1} \text{ mm}^{-2}$ also identified a similar phenomenon [8]. The advantage of the results presented here are the high sensitivity of the measurements and the ability to study the phenomenon’s dynamics on a $< 1 \text{ ms}$ time scale.

4.2.2 Time dependence

A series of dynamic datasets were collected following the identification of the ‘excess leakage-current effect’, during which experimental conditions were varied over an extended period ($> 10 \text{ s}$), to further investigate the processes governing this phenomenon.

One 20 s dataset, collected at 20 keV in the low-gain stage, comprised cycling the opening and closing of the JJ-slits used to define the beam size. A schematic of this measurement is shown in figure 8(a), and results obtained for pixel (13,33) are presented in figure 8(b). Here, the shift in the 0 γ peak position relative to a dark dataset is plotted over a 10^4 frame (10 ms) window (blue) alongside an estimate of the interacting X-ray flux within the pixel (red). The evolution of the estimated flux (section 3.1) indicates that pixel (13,33) was first exposed to radiation after $\sim 4.50 \text{ s}$ and that the beam size was subsequently cycled twice fully and opened a third time after $\sim 18.27 \text{ s}$. A constant 40 mm of Al attenuation was used throughout the capture and, when exposed, an X-ray flux of $\sim 1.18 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ (7.34 % occupancy) was applied to the pixel.

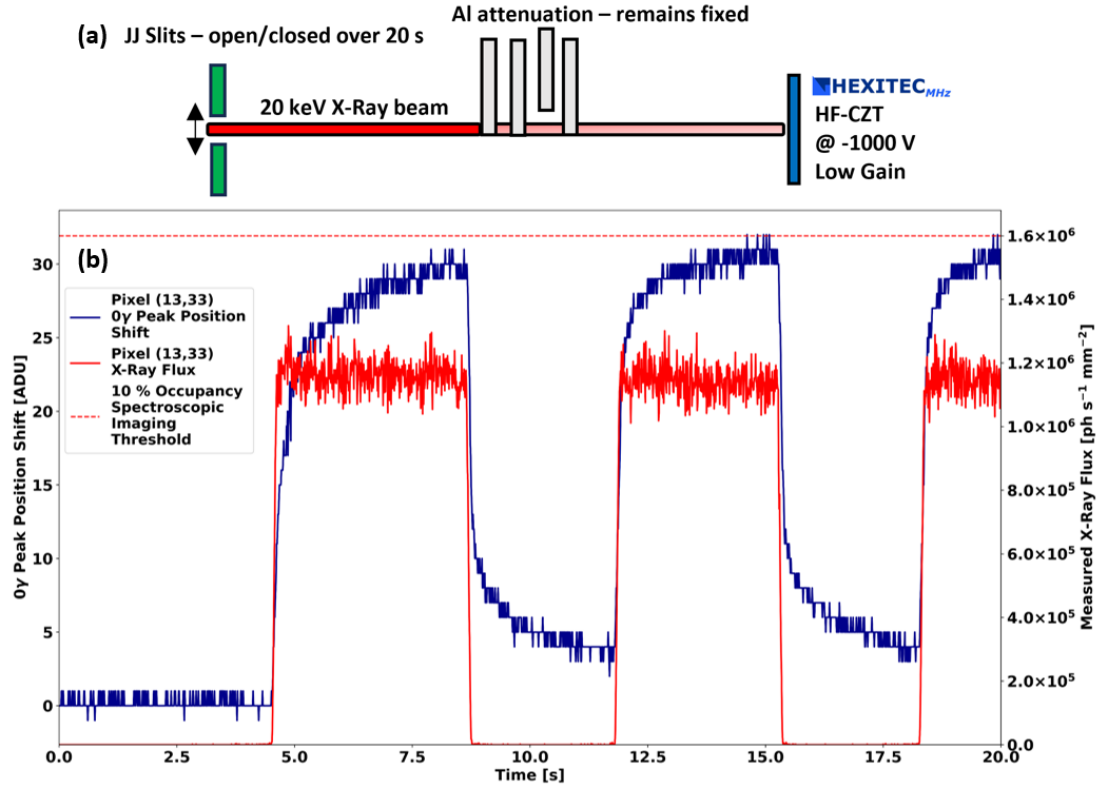


Figure 8. Results from a 20 s low-gain dataset collected at 20 keV whilst cycling the JJ slits. (a) Schematic of measurement protocol. (b) The evolution of the average 0 γ peak position (blue) and estimated interacting X-ray flux (red) for pixel (13,33) over 20 s — 10^4 frame sampling window.

Prior to exposure, displacements in the 0γ peak were negligible and the small 1 ADU fluctuations were at the noise level of the sensor.

An immediate shift of 11 ADU was measured upon irradiation, and its magnitude continually rose throughout the duration of the pixel's ~ 4.16 s initial exposure. During this period, the peak moved by a further 20 ADU, resulting in a final displacement of 31 ADU relative to the dark dataset. This is equivalent to a shift of ~ 3.07 keV in the low-gain operating mode. However, the curvature of the plot at the instant of slit closure (~ 8.66 s) indicates that the offset had not yet stabilised, and a steady state had not yet been achieved. This is further evidenced by the second exposure, where a final shift of 32 ADU was recorded following a shorter ~ 3.47 s irradiation.

Achievement of a larger offset following a shorter exposure was the consequence of the offset's decay following a reduction in the applied flux. Although 50 % of the shift was recovered in the initial ~ 0.09 s, the pixel's 0γ offset returned only to a value of 4 ADU during the ~ 3.15 s separating the closing and reopening of the beam.

Additional datasets are required to better understand the precise dynamics of this phenomenon. For example, longer exposures are needed to ascertain whether a steady-state leakage current is eventually achieved following the application of a specific flux. Additionally, a prolonged collection following the removal of an applied flux would allow better quantification of the offset's decay and should track the detector's behaviour until it returns to the original, unexposed state.

These datasets demonstrate the capability of $\text{HEXITEC}_{\text{MHz}}$ to track such material effects on a per-pixel basis and MHz timescale. Crucially, the implementation of a Xilinx[®] Alveo U50 Data Center Accelerator card as a second-stage FPGA (section 2.2) will enable such analysis to be performed during capture. For typical spectroscopic applications, $\text{HEXITEC}_{\text{MHz}}$ will operate at fluxes of $\sim 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$, where the observed ~ 3 keV shifts in the pixel output can be tracked and corrected for in real time by the FPGA. Hence, a typical end-user of HF-CZT $\text{HEXITEC}_{\text{MHz}}$ devices would be unaffected by this phenomenon.

4.2.3 Intensity dependence

Figure 9(a) shows a schematic for an additional dataset taken to investigate the dynamics of the 'excess leakage-current effect', in which Al attenuation was cycled in and out of the 20 keV X-ray beam over 20 s. As in section 4.2.2, the evolution of the flux applied to pixel (13,33) and its corresponding 0γ peak position shift is plotted as a function of time in figure 9(b). Starting with 7.0 mm of Al attenuation, the X-ray shutter was opened after ~ 2.64 s and Al gradually removed and subsequently re-inserted into the beam (steps of 7.0 mm, 6.5 mm, 5.5 mm, 3.5 mm, 1.50 mm) prior to the closing of the shutter after ~ 19.01 s. When exposed, attenuation cycling resulted in the application of measured X-ray fluxes of $\sim 8.12 \times 10^4 \text{ ph s}^{-1} \text{ mm}^{-2}$, $\sim 1.24 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$, $\sim 3.10 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$, $\sim 1.85 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$ and $\sim 9.60 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$.

These results corroborate the findings of section 4.2.2 and the application of a specific X-ray flux resulted in a continual rise of the 0γ peak position shift, indicative of an associated rise in the measured leakage current towards a supposed steady state. This was not achieved at the higher X-ray fluxes utilised, and the small offsets present at the lower fluxes make determination of a steady state difficult due to the maximum 1 ADU resolution.

The dataset also provides support for the hypothesis that the magnitude of the 'excess leakage-current effect' is dependent on the X-ray flux, with larger fluxes resulting in greater shifts in the

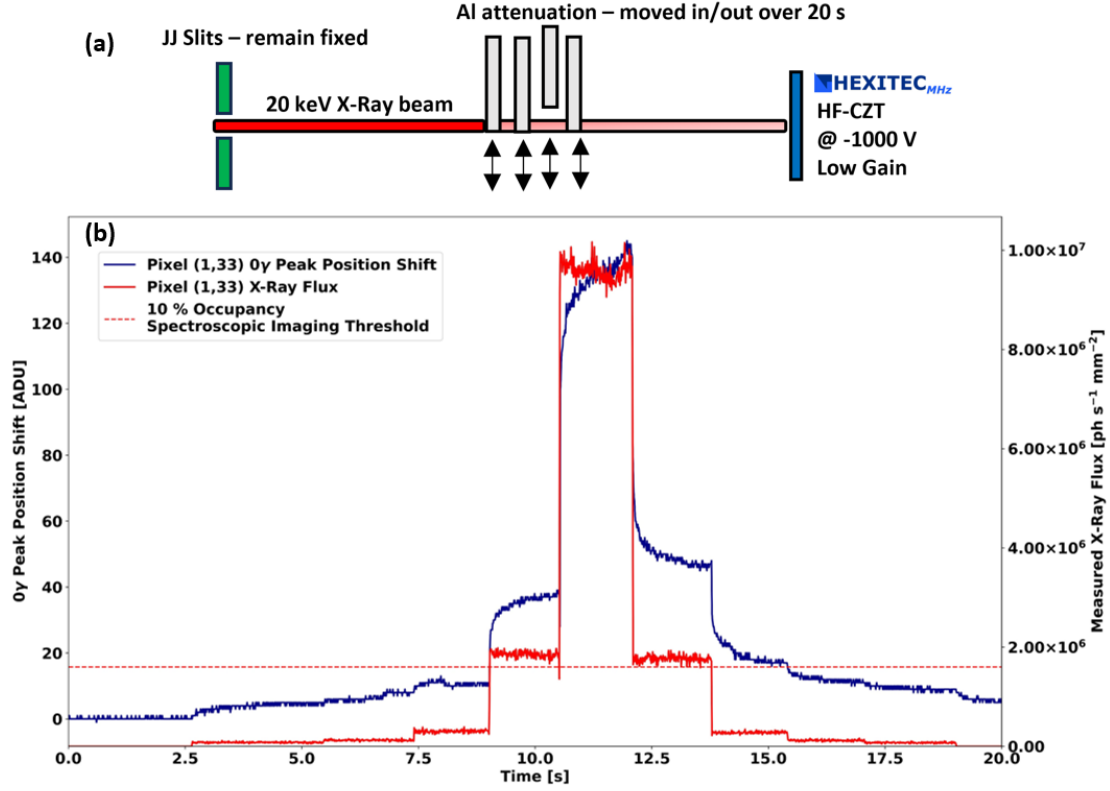


Figure 9. Results from a 20 s low-gain dataset collected at 20 keV whilst cycling the Al attenuators. (a) Schematic of measurement protocol. (b) The evolution of the average 0γ peak position shift (blue) and measured X-ray flux (red) for pixel (13,33) over 20 s — 10^4 frame sampling window.

0γ peak position. The lack of steady-state attainment, due to the short exposure time at each attenuation level, makes a quantitative analysis of the correlation between the X-ray flux and the effect's magnitude difficult. However, a preliminary assessment of this relationship for pixel (13,33) is shown in figure 10(a). Here, the offsets used represent the final offset measured at each attenuation level during the initial ~ 12.05 s, whilst Al was gradually removed from the beam. Additionally, values for the X-ray flux at each attenuation level have been calculated through extrapolation of the intensity measured with 7 mm of Al, using the literature 20 keV Al attenuation coefficient [30]. The deviations between these calculated intensities and those measured on HEXITEC_{MHz} are shown in figure 10(b) and are the consequence of a breakdown at higher occupancies in the assumption that HEXITEC_{MHz} operates as a nonparalyzable counting system (section 3.1).

For the attenuation thicknesses used, offsets of 5 ADU, 8 ADU, 11 ADU, 38 ADU and 144 ADU were measured respectively. Using the 99.0 eV ADU^{-1} low-gain gradient calculated (section 3.2), the 0γ peak offset of 144 ADU measured at a calculated flux of $1.26 \times 10^7 \text{ ph s}^{-1} \text{ mm}^{-2}$ is equivalent to 14.26 keV. This represents 4.75 % of the $\sim 300 \text{ keV}$ HEXITEC_{MHz} low-gain dynamic range. Equivalently, this value corresponds to a per-pixel leakage current of $\sim 543 \text{ pA}$ (8.68 nA mm^{-2}), assuming a CZT electron-hole-pair generation energy of 4.67 eV [28]. Although all pixels displayed shifts within the same order of magnitude, pixel-to-pixel performance differences were seen. This is a consequence of pixel-wise and sub-pixel variations within the device and will be investigated further when a 20-channel readout is available (section 3.1).

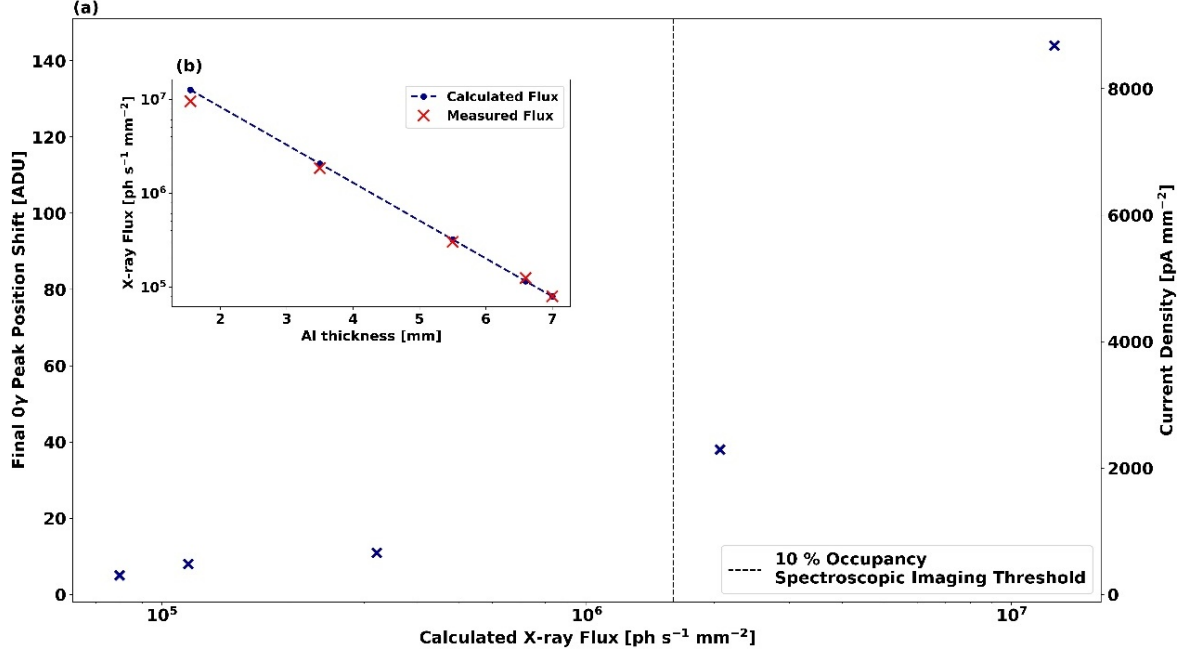


Figure 10. Results from a 20 s low-gain dataset collected at 20 keV whilst cycling the Al attenuators. (a) The 0 γ peak position shift at each attenuation level as a function of the calculated X-ray flux (red) for pixel (13,33). The vertical dashed line marks the maximum flux used for spectroscopic measurements with HEXITEC_{MHz}. (b) A comparison of the calculated and measured X-ray fluxes for the different thicknesses of Al attenuation used.

However, fluences of $10^7 \text{ ph s}^{-1} \text{ mm}^{-2}$ represent single-pixel occupancies of 62.5 %, far above the 10 % upper limit for typical spectroscopic measurements (section 3.2). This corresponding flux, $1.60 \times 10^6 \text{ ph s}^{-1} \text{ mm}^{-2}$, is marked by the vertical dashed line in figure 10(a). Here, the relative offsets seen are significantly lower and spectroscopic measurements are typically made far below the supposed limit (at occupancies of ~ 2 %). At a calculated flux of $3.21 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$ (~ 2 % occupancy), the 0 γ peak position of pixel (13,33) is shifted by 11 ADU (663 pA mm⁻²). Any shifts of this magnitude will be corrected for using real-time FPGA processing.

4.2.4 Spatial distribution

An additional fundamental property of the ‘excess leakage-current effect’ is its spatial distribution across the active area.

Upon identifying the effect at the B16 Test Beamline, results were compared to laboratory measurements taken using a continuous Comet MXR-160/22 Industrial X-ray source. Here, the identical HF-CZT HEXITEC_{MHz} device was mounted in the direct beam, with a line pair per mm slide placed over the active area to generate regions of higher and lower X-ray occupancy (figure 11). The FPGA firmware for readout of the high-speed serialisers (section 2.2) was unavailable at the time of these measurements and these tests therefore utilised the SPI interface for readout of the TDCs [15]. The 80×80 mappings shown in the results were generated by cycling the sector sampled across the active area.

Figure 12 shows a high-gain dataset obtained using an 80 kV tube voltage and 2 mA tube current. These settings resulted in a polychromatic tube output of mean energy ~ 23 keV, with an average

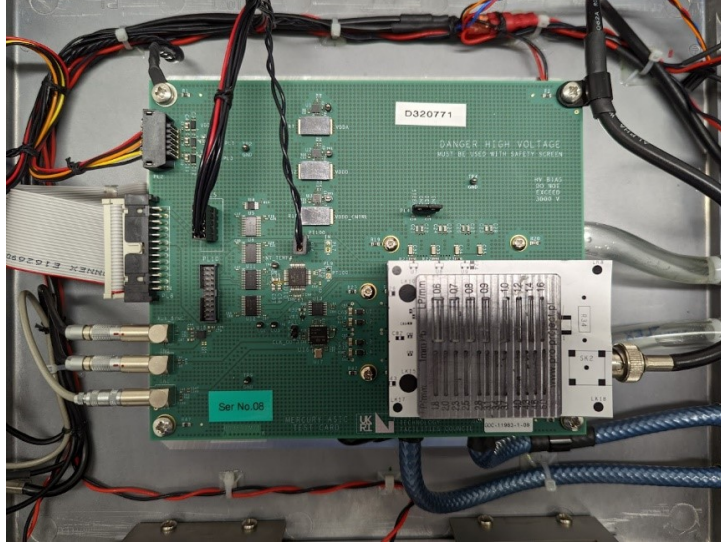


Figure 11. A lead line pair per mm slide placed over the HF-CZT HEXITEC_{MHz} device within the test enclosure.

occupancy of $\sim 1.94\%$ ($\sim 3.16 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$) recorded across the 80×80 array. However, as shown in figure 12(a), regions of differing occupancy resulted from the geometry and design of the Pb line pair per mm slide. For example, pixel (60,40), situated in an area of lower attenuation, experienced an applied flux of $\sim 6.70 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$. Conversely, the more-heavily-shielded pixel (40,66) faced a lower flux of $\sim 0.68 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$. These gradients in the applied flux strongly correlate to the shifts measured in each pixel's 0γ peak relative to a dark dataset (Fig 12b). In general, higher-occupancy pixels displayed greater shifts, with many pixels in highly-exposed regions showing offsets > 50 ADU, compared to shifts < 10 ADU in those more highly-attenuated. This is shown more clearly in the three dark-corrected histograms of figure 12(c). These chosen pixels displayed shifts of 6 ADU, 23 ADU and 61 ADU at respective fluxes of $\sim 0.68 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$, $\sim 5.23 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$ and $\sim 6.79 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$. Overall, these correlations match earlier observations and the strong replication of the slide's geometric features in Fig 12c suggest that the 'excess leakage-current effect' is highly localised, with pixels affected little by the radiation activity of their nearest neighbours.

To test this latter hypothesis quantitatively, datasets were collected at DLS in which the Front End's CSA reset was impeded for $10 \mu\text{s}$, the equivalent of ten 1 MHz frame periods. The ASIC continued to output data at a MHz rate when operated in this manner, allowing measurement of the integrated charge within each pixel for ten integration periods ($0.9 \mu\text{s}$, $1 \mu\text{s}$, $2 \mu\text{s}$, $3 \mu\text{s}$... $10 \mu\text{s}$). The first integration period ($0.9 \mu\text{s}$) is shorter than the $1 \mu\text{s}$ interval between subsequent integration lengths on account of the 100 ns reset at the beginning of each 10 frame bunch. The collation of equivalent frames enabled a histogram to be generated for each integration length. This data was subsequently used to estimate a per-pixel leakage current through identification of the position of the 0γ peak within each distribution. These ADU values were converted into an equivalent charge using the CZT electron-hole-pair generation energy (4.67 eV [28]) and the ADU-keV conversion factors calculated during calibration (section 3.2). The gradient of a linear fit to a plot of the integrated charge against integration time represents the estimated leakage current.

The aforementioned process is shown for a single pixel in figure 13. The corresponding high-gain dataset was collected using a $1 \text{ mm} \times 2.5 \text{ mm}$ 60 keV beam and attenuation comprising 0.5 mm of

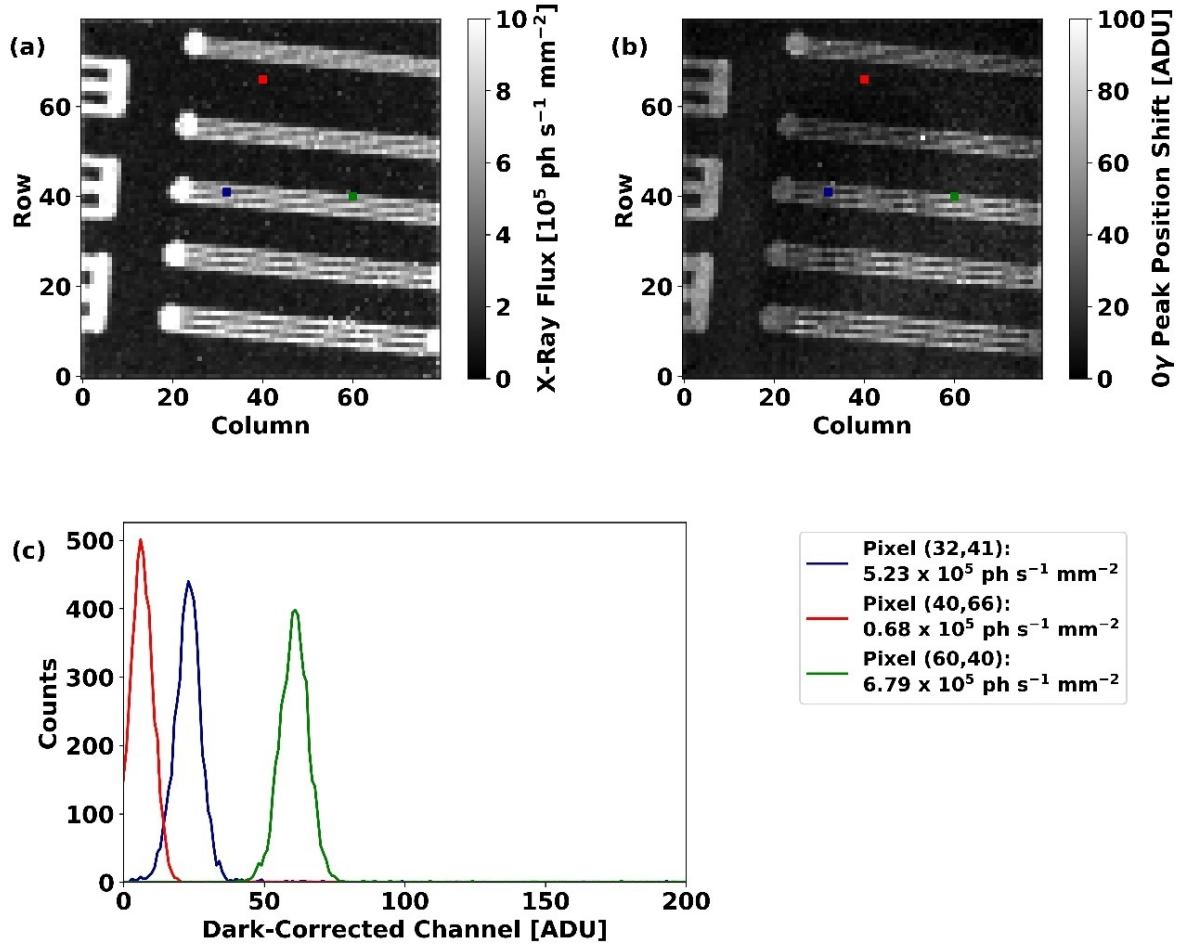


Figure 12. High-gain results from laboratory direct X-ray illumination measurements of a lead line pair per mm slide at 80 kV, 2 mA. (a) Map of the interacting X-ray flux. (b) Map of the 0γ peak position shifts relative to the dark dataset. (c) Dark-corrected histograms of the signals measured in three pixels — 1 ADU bin width. Pixels shown in (c) are highlighted in (a) and (b).

stainless steel and 2 mm of Al. This resulted in an average occupancy within the 4×10 pixel irradiated area of $\sim 4.37\%$ ($\sim 6.99 \times 10^5 \text{ ph s}^{-1} \text{ mm}^{-2}$). In order to allow the subsequent comparison of all read-out pixels, the average $37.04 \text{ eV ADU}^{-1}$ conversion factor of irradiated pixels was used for all 320 channels. A per-pixel calibration was not possible for the unirradiated channels since the 1γ peak is absent from the raw spectra (section 3.2).

Examination of these results globally across the serialiser channel, as shown in figure 14, allows the spatial distribution of the ‘excess leakage-current’ effect to be assessed. Figure 14(a) shows the linear fits generated across all 320 channels, with red and blue lines used to represent irradiated and unirradiated pixels respectively. The single green plot corresponds to pixel (13,65), a defective pixel resulting from an imperfection in the hybridisation process (section 2.3). The clear separation of the red and blue distributions indicates distinct differences between pixels inside and outside of the beam, with irradiated pixels displaying notably higher leakage currents (gradients). In fact, the mean leakage current of irradiated pixels was 31.48 pA (504 pA mm^{-2}), compared to the 9.51 pA (152 pA mm^{-2}) value calculated for unirradiated channels. This ‘excess leakage current’ of 352 pA mm^{-2} is reflected

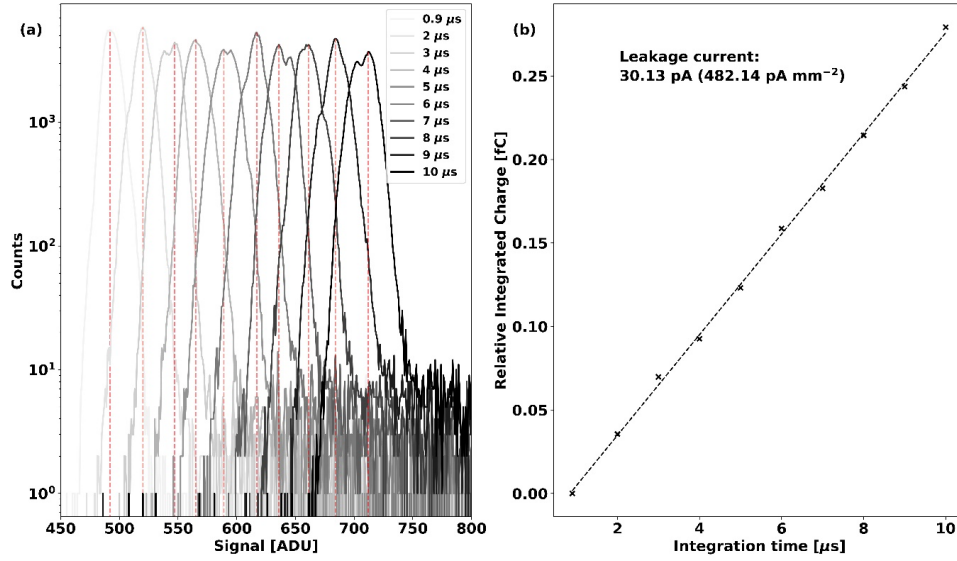


Figure 13. Results from a 60 keV high-gain integration-length scan for pixel (14,40). (a) The raw histograms associated with each integration length — 1 ADU bin width. The positions of the 0 γ peak in each histogram are marked by the dashed red lines. (b) The linear fit generated from a plot of the integration time against the integrated charge equivalent to the 0 γ peak’s centroid.

in the leakage current map shown in figure 14(b). As above, these results imply that the ‘excess leakage-current effect’ is highly localised. This is further evidenced by the currents calculated for pixels (14,42) and (14,43), which span the exposed area’s boundary. Here, the irradiated pixel (14,42) showed a measured current of 30.29 pA (485 pA mm⁻²), 21.08 pA (337 pA mm⁻²) higher than the 9.21 pA (147 pA mm⁻²) value calculated for its unirradiated neighbour.

For comparison, the aforementioned characterisation of HF-CZT using electroless gold and sputtered platinum electrodes at the Extremely Brilliant Source of the ESRF reported excess leakage currents of ~ 10 pA at fluxes of $\sim 10^7$ ph s⁻¹ mm⁻² [8]. With pixel dimensions of 500 μ m $\times$ 500 μ m, this is equivalent to a current density of ~ 40 pA mm⁻². The values reported here therefore greatly exceed those previously measured and may be explained by the differences in either the contact technologies or the interconnect processes used for a simple planar device and the HEXITEC_{MHz} pixelated sensor. However, further datasets are required to fully investigate these differences.

5 Conclusion

A 2 mm thick HF-CZT sensor, manufactured by Redlen Technologies, was hybridised to the HEXITEC_{MHz} ASIC and characterised at the DLS synchrotron’s B16 Test Beamline using a uniform and monochromatic beam. The results obtained were presented.

Under spectroscopic X-ray fluxes ($< 1.6 \times 10^6$ ph s⁻¹ mm⁻²), resolutions of 850 ± 100 eV, 920 ± 100 eV and 1130 ± 120 eV were measured at 20 keV in the high-, medium- and low-gain operating stages respectively. These results exceed the targeted < 1 keV resolution of the medium-gain stage.

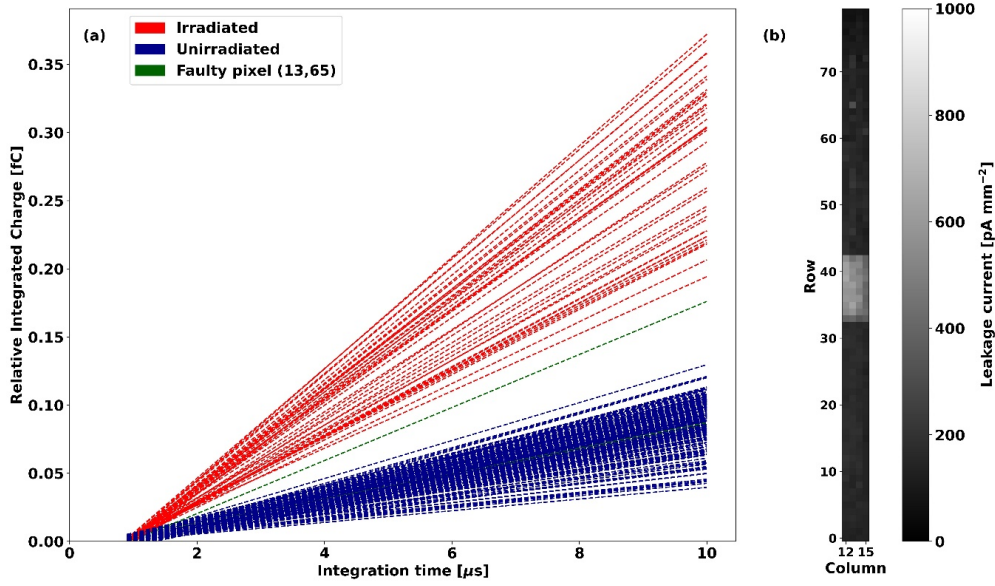


Figure 14. Results from a 60 keV high-gain integration-length scan. (a) The linear fits generated from plots of the integration time against the integrated charge equivalent to the 0γ peak’s centroid for all pixels in the single 4×80 channel. Fits associated with pixels inside and outside the irradiated area are coloured red and blue respectively. The fit associated with the faulty pixel (13,65) is highlighted in green. (b) A map of the calculated leakage currents (i.e. fit gradients) across the 4×80 channel.

Measurements also indicated a phenomenon whereby excess leakage current was generated in the detector following the initial application of a specific X-ray flux. This manifested in a temporary shift in raw spectra to higher ADU values, shown throughout these results in the changing position of the 0γ peak. Datasets taken to analyse the effect’s dynamics showed it to be highly localised, with pixels unaffected by the irradiation of their nearest neighbours, and intensity-dependent. For example, at an X-ray flux of $1.26 \times 10^7 \text{ ph s}^{-1} \text{ mm}^{-2}$, a characteristic pixel displayed an offset equivalent to 14.26 keV or 543 pA (8.68 nA mm^{-2}). Comparison to p-type Si HEXITEC_{MHz} data suggests this is a CZT-specific phenomenon and it is hypothesised that it results from charge trapping at the electrode-CZT interface. This results in the localised build-up of space charge and modification of the contact potential barrier. Further datasets are required to fully test this hypothesis and develop more quantitative measures of the buildup and decay of leakage current and to establish whether a steady-state is eventually achieved.

The ‘excess leakage-current effect’ and its consequences clearly represent a complex problem that is of relevance to the entire photon-science community. A collaborative effort on multiple fronts is required to further investigate its origin and any potential solutions. Future work at RAL will look to build upon this initial analysis, facilitated by the delivery of the FPGA firmware for a full 80×80 readout which will enable an in-depth investigation into variations seen across the active area.

Importantly, the identification and subsequent analysis of this phenomenon also demonstrates the capacity of HEXITEC_{MHz} to track and diagnose such material effects on a MHz timescale. This capability will also enable a real-time correction of the effect, facilitated by the implementation of a Xilinx[®] Alveo U50 Data Accelerator card as a second-stage FPGA.

Acknowledgments

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